**3.** **b.**

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ -1 & 1 & 2 & b_2 \\ 1 & 3 & 4 & b_3 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 2 & 3 & b_1 + b_2 \\ 0 & 2 & 3 & b_3 - b_1 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 2 & 3 & b_1 + b_2 \\ 0 & 0 & 0 & b_3 + 3b_1 \end{array}\right]$$

So the constraint equation is $2b_1 + b_2 - b_3 = 0$.

**4.** To be an element of $V$ we must solve $A\mathbf{x} = \mathbf{b}$.

**a.**

$$\left[\begin{array}{cc|c} -1 & 2 & b_1 \\ 2 & -4 & b_2 \\ 1 & -2 & b_3 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -2 & -b_1 \\ 0 & 0 & b_2 + 2b_1 \\ 0 & 0 & b_3 + b_1 \end{array}\right]$$

So then $\mathbf{b}$ must satisfy the constraint equations $b_2 + 2b_1 = 0$ and $b_3 + b_1 = 0$.

**c.**

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & 1 & -1 & b_2 \\ 1 & 1 & 1 & b_3 \\ 1 & 2 & 0 & b_4 \end{array}\right] \rightsquigarrow \left[\begin{array}{cccc} 1 & 0 & 2 & b_1 \\ 0 & 1 & -1 & b_2 \\ 0 & 0 & 0 & b_3 - b_3 - b_1 \\ 0 & 0 & 0 & b_4 - 2b_2 - b_1 \end{array}\right]$$

So then $\mathbf{b}$ must satisfy the constraint equations $b_1 + b_2 - b_3 = 0$ and $b_1 + 2b_2 - b_4 = 0$.

**5** Rewrite the parametric equation as a matrix $[A\mathbf{x}|\mathbf{b}]$. The Cartesian equation will then be the constraint equation.

**a.**

$$\left[\begin{array}{cc|c} 1 & 2 & b_1 \\ -2 & 0 & b_2 \\ -2 & -1 & b_3 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 0 & 4 & b_2 + 2b_1 \\ 0 & 0 & 2b_1 - 3b_3 + 4b_3 \end{array}\right]$$

So the constraint equation give the Cartesian equation, $2x_1 - 3x_3 + 4x_3 = 0$.

**10.** We can give a counterexample which proves that the statement is false. Let

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

so that $A\mathbf{x} = \mathbf{0}$ certainly has the solution $\mathbf{x} = \mathbf{0}$, but

$$A\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

has no solution.

**15.** **a.** $\mathbf{a_1} + \mathbf{a_2} + \cdots \mathbf{a}_n = \mathbf{0}$ implies that $A\mathbf{x} = \mathbf{0}$ has the nontrivial solution $\mathbf{x} = (1, 1, \ldots, 1)$ where $\mathbf{x} \in \mathbb{R}^n$. Thus by Proposition 5.5 $A$ is singular which by definition means that $\text{rank}(A) < n$.

**b.** As above, if there is a nonzero vector $\mathbf{c} = (c_1, c_2, \ldots, c_n)$ such that $\sum_{i=1}^{n} c_i\mathbf{a}_i = \mathbf{0}$, then $A\mathbf{x} = \mathbf{0}$ has the nontrivial solution $\mathbf{x} = \mathbf{c}$ implying that $A$ is singular and that $\text{rank}(A) < n$.